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CT PERFORMANCE EVALUATION REPORT

PRE-TEST / POST FAN CHANGE / POST DISTRIBUTION CHANGE — THREE-STAGE SINGLE CELL ASSESSMENT

CT - 02 , CELL 02

OWNER / CLIENT	**DHARIWAL INFRASTRUCTURE LTD CHANDRAPUR**
TESTING AGENCY & EQUIPMENT PROVIDER	**SS COOLING TOWER CONSULTANTS (SSCTC)**
DATE OF TEST	**27 March 2026**
REPORT DATE	**15 April 2026**
ASSESSMENT METHOD	**CTI ATC-105 Merkel Method — Applied independently to each of 3 test stages**

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PREAMBLE

Due to poor performance of the cooling tower, factors influencing the poor performance were identified to be progressively resolved. To carry out this procedure it was decided to carry out the changes on one cell and based on the result carry out the same on balance cell. This procedure had to be conducted both before change of gear box and fan and post change of gear box and fan to ascertain the effect on cold water temperature.

MEMBERS PRESENT FOR THE TEST

CLIENT — DHARIWAL INFRASTRUCTURE LTD CHANDRAPUR

- Mr Shrikant Shrivastava (Remote)
- Mr GAURAV HANTODKAR

- Mr PAWAN GAWANDE

SS COOLING TOWER CONSULTANTS (SSCTC)

- Mr SURESH SARMA
- Mr SANJAY GORAD
- Mr MRADUL VISHWAKARMA
- Mr PARAG VISHWAKARMA
- Mr RAHUL MANKE

ASSESSMENT METHOD & ANALYSIS

Pre test was conducted on 27 March 2026.

Cell has been isolated at the basin level for collection of cold water directly from the rain zone.

All required data was collected and reports have been prepared for each test stage.

Since different conditions apply in each test, both pre and post tests have been compared to the design conditions. The difference between pre and post tests has been established accordingly.

INSTRUMENT PLACEMENT AND READINGS

Air flow was measured using Data Logging anemometer as per CTI ATC-143 Method of equal area — 10 traverses per quadrant (40 readings total for one fan).

Hot water temperature was taken at inlet of the hot water to the cooling tower.

Cold water temperature: 24 RTD sensors placed at air-inlet sides of the isolated cell (12 per side).

Water flow was measured using UFM (GE Make) on the riser.

WBT/DBT was measured using wet-bulb automatic stations recording every minute.

Power to the fan motor was noted from the client MCC.

CONCLUSION AND REMARKS

1. Testing of the cell was carried out after incorporating the distribution system changes on 27th of March 2026.
2. The additional head available caused flow to increase substantially. Even under these conditions the cell showed an improvement of approx. 0.5 °C above the post fan modification test.
3. Single cell isolated testing involves UNCERTAINTY in measurement due to turbulent water flow.

SUGGESTIONS AND SPECIAL NOTES

1. Improvement of 1.5 appears high in "TEST 2" so we consider that as 0.7 Deg C
2. Single cell isolated testing involves UNCERTAINTY in measurement due to turbulent water flow

FINAL DATA AND CONCLUSION OF PRE AND POST TEST

All three tests have been independently evaluated against the same Design baseline using the CTI ATC-105 five-step method. CWT Deviation from Design and Capability are computed values, not raw measurements.

Parameter	Unit	TEST 1 Pre-Test	TEST 2 Post Fan	TEST 3 Post Distr.
Water Flow	M3/hr	2998	3067.21	**3680**
WBT	Deg.C	25.25	24.22	**21.7**
HWT	Deg.C	44.67	43.21	**42.13**
CWT	Deg.C	35.08	32.89	**32.4**
Fan Power At Motor Inlet	KW	97.04	116.24	**117**
Range	Deg.C	9.59	10.32	**9.73**
Approach	Deg.C	9.83	8.67	**10.7**
CWT Deviation from Design	Deg.C	-14.82	-17.60	**-22.11**
Capability	%	395.1	373.2	**359.0**
Improvement vs Previous Test	Deg.C	—	2.77	**4.51**
Cumulative Improvement vs Test 1	Deg.C	—	2.77	**7.28**

Notes / Mathematical Findings:

1. Improvement of 1.5 appears high in "TEST 2" so we consider that as 0.7 Deg C
2. Single cell isolated testing involves UNCERTAINTY in measurement due to turbulent water flow

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ATC-105 PERFORMANCE EVALUATION — TEST 1 — PRE TEST

Parameter	Unit	Design	Recorded Test
Water Flow Rate	M3/hr	1000	2998
Hot Water Temperature	Deg C	37	44.67
Cold Water Temperature	Deg C	32	35.08
Wet Bulb Temperature	Deg C	27	25.25
Range (HWT – CWT)	Deg C	5	9.59
Approach (CWT – WBT)	Deg C	5.0	9.83

STEP 1

Table 1 — CWT vs Range at Test WBT = 25.25°C · Design Range = 5°C

Performance curves at 90%, 100% and 110% water flow are evaluated at Test WBT = 25.25°C. CWT values at 80%, 100% and 120% of design range are computed for each flow condition.

Range %	Range °C	CWT 90% Flow (900)	CWT 100% Flow (1000)	CWT 110% Flow (1100)
80%	4 °C	29.00	29.43	29.85
100%	**5 °C**	**29.73**	**30.23**	**30.71**
120%	6 °C	30.38	30.94	31.48

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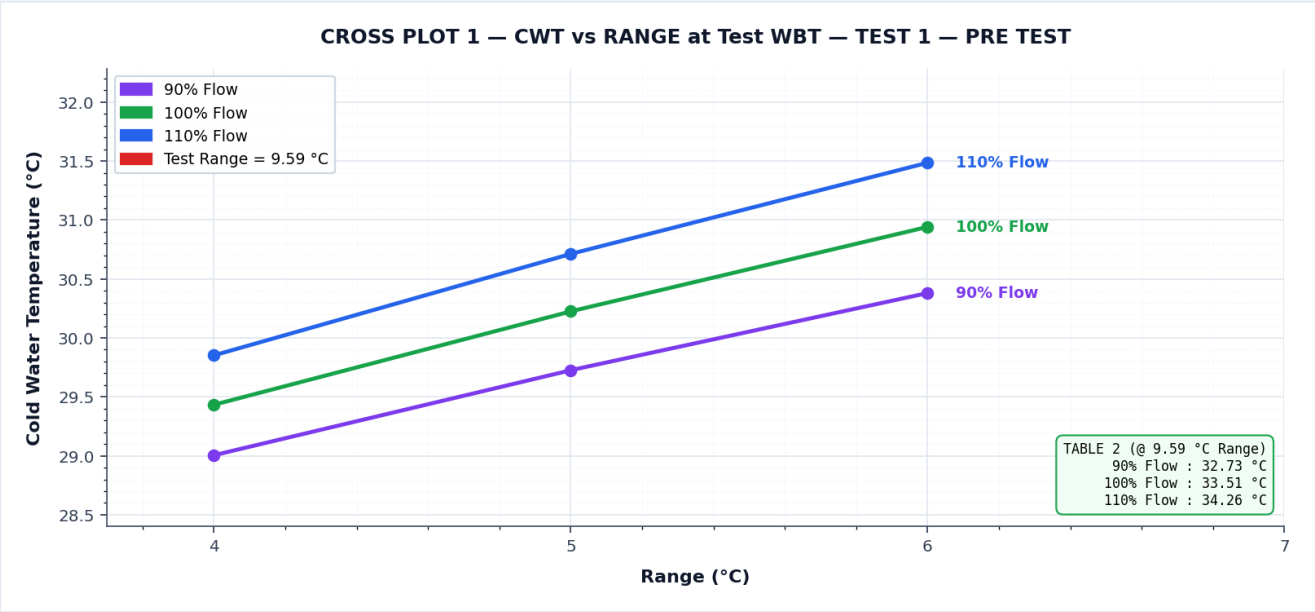
STEP 2

Cross Plot 1 — CWT vs Range at Test WBT = 25.25°C [TEST 1 — PRE TEST]

CWT (Y-axis) vs Range (X-axis) at 90%, 100% and 110% water flow, evaluated at Test WBT = 25.25°C. Draw a vertical line at test range 9.59°C to read the CWT at each flow — these three intersection points form Table 2 and are used in Cross Plot 2.



CROSS PLOT 1 [TEST 1 — PRE TEST] — CWT vs RANGE at Test WBT = 25.25°C · Test Range (red vertical line) = 9.59°C

TABLE 2 — CWT at Test Range = 9.59°C (Read from Cross Plot 1)

Flow Condition	Water Flow (m3/hr)	CWT at Test Range (°C)
90% Flow (Purple)	900	32.728
100% Flow (Green)	1000	33.511
110% Flow (Blue)	1100	34.26

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STEP 3

Calculation of Adjusted Water Flow (Q_{adj}) [TEST 1 — PRE TEST]

Correct the test water flow for fan power and air density differences between design and test conditions. The adjusted flow is then marked on Cross Plot 2.

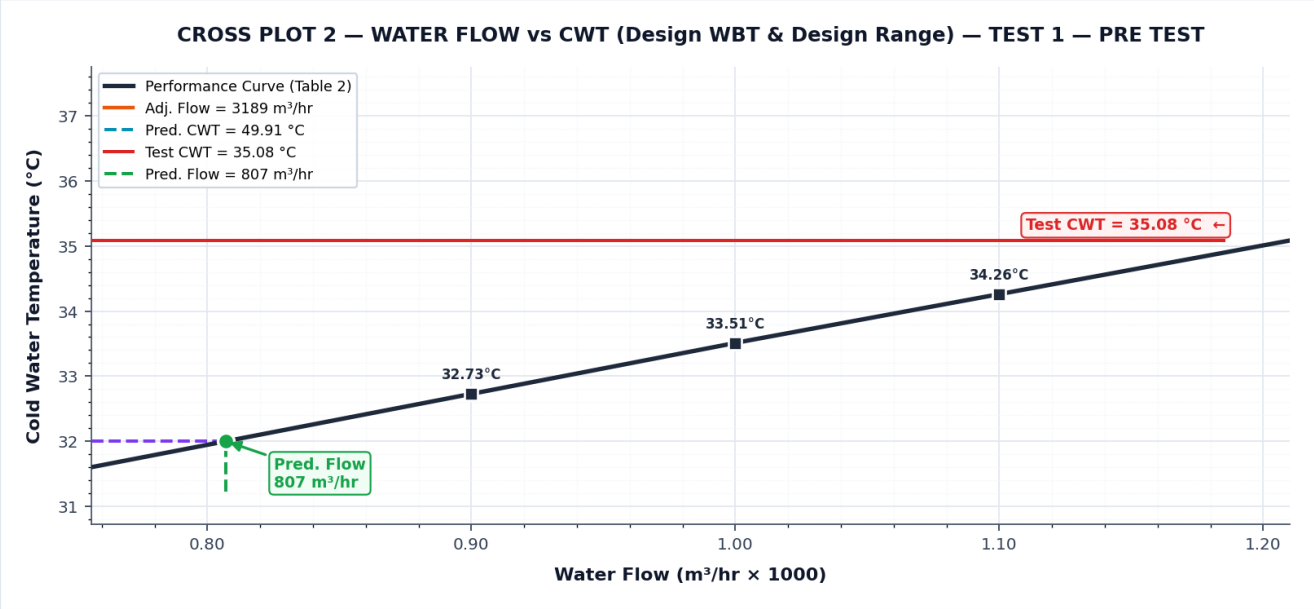
Step 3 — Input Parameters & Adjusted Flow Calculation

Ref	Parameter	Value	Unit
a	**Fan Power Design (W_d)**	117	kW
b	**Fan Power Test (W_t)**	97.04	kW
c	**Hot Water Temperature (Test)**	44.67	Deg C
d	**Cold Water Temperature (Test)**	35.08	Deg C
e	**Range (Test)**	9.59	Deg K
f	**WBT (Test)**	25.25	Deg C
g	**Approach (Test)**	9.83	Deg C
h	**Test Water Flow (Q_{wt})**	2998	M3/hr
i	**Density Ratio (test / design)**	0.998049	—
$Q_{adj} = Q_{wt} \times (W_d/W_t)^{1/3} \times (\text{density ratio})^{1/3}$		**3188.80**	**m3/hr**

STEP 4

Cross Plot 2 — Water Flow vs CWT [TEST 1 — PRE TEST]

Plot the three Table 2 points (Water Flow on X-axis, CWT on Y-axis). Mark Q_{adj} on the X-axis and project vertically to the curve to read the Predicted CWT. Project horizontally at Design CWT (32°C) to read the Predicted Water Flow.



CROSS PLOT 2 [TEST 1 — PRE TEST] — Water Flow vs CWT · Adjusted Flow = 3188.80 m3/hr · Predicted CWT = 49.91°C

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FINAL ATC-105 RESULTS — TEST 1 — PRE TEST

STEP 5 **Predicted CWT, Performance Shortfall and Capability — TEST 1 — PRE TEST**

Mark the Adjusted Water Flow on the X-axis of Cross Plot 2 and project vertically to intersect the performance curve. Read the Predicted CWT horizontally from that point.

Ref	Item	Value	Unit
i	**Adjusted Water Flow (X-axis of Cross Plot 2)**	3188.80	m3/hr
ii	**Project vertical line → intersect curve → read CWT**	See Cross Plot 2 — orange vertical line at adjusted flow	
iii	**Predicted (Expected) Cold Water Temperature**	**49.91**	**Deg C**
iv	**Tolerance on CWT (Specification)**	0.00	Deg C
v	**Total Predicted CWT (incl. tolerance)**	**49.91**	**Deg C**
vi	**Test Cold Water Temperature (Recorded site data)**	35.08	Deg C
vii	**Predicted Water Flow (at Design CWT on Cross Plot 2)**	807.02	m3/hr

CONCLUSION — TEST 1 — PRE TEST

CWT Shortfall in Performance	Cooling Tower Capability
-14.82 °C	**395.1 %**
Test CWT (35.08°C) vs Predicted CWT (49.91°C)	= 3188.80 / 807.02 × 100

[TEST 1 — PRE TEST] — Shortfall = **-14.82 °C** · Capability = **395.1 %**

Test CWT is -14.82°C above the Predicted CWT. Capability = Adjusted Flow / Predicted Flow × 100 = 3188.80 / 807.02 × 100.

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ATC-105 PERFORMANCE EVALUATION — TEST 2 — POST FAN CHANGE

Parameter	Unit	Design	Recorded Test
Water Flow Rate	M3/hr	1000	3067.21
Hot Water Temperature	Deg C	37	43.21
Cold Water Temperature	Deg C	32	32.89
Wet Bulb Temperature	Deg C	27	24.22
Range (HWT – CWT)	Deg C	5	10.32
Approach (CWT – WBT)	Deg C	5.0	8.67

STEP 1

Table 1 — CWT vs Range at Test WBT = 24.22°C · Design Range = 5°C

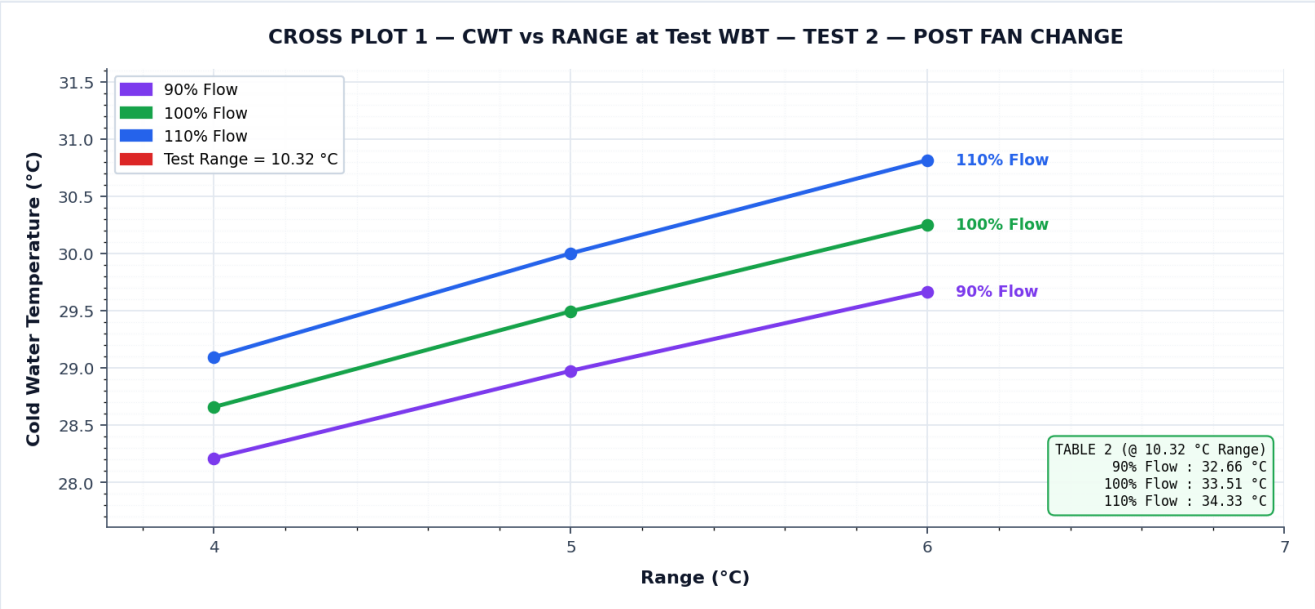
Performance curves at 90%, 100% and 110% water flow are evaluated at Test WBT = 24.22°C. CWT values at 80%, 100% and 120% of design range are computed for each flow condition.

Range %	Range °C	CWT 90% Flow (900)	CWT 100% Flow (1000)	CWT 110% Flow (1100)
80%	4 °C	28.21	28.66	29.10
100%	**5 °C**	**28.98**	**29.50**	**30.00**
120%	6 °C	29.67	30.25	30.82

STEP 2

Cross Plot 1 — CWT vs Range at Test WBT = 24.22°C [TEST 2 — POST FAN CHANGE]

CWT (Y-axis) vs Range (X-axis) at 90%, 100% and 110% water flow, evaluated at Test WBT = 24.22°C. Draw a vertical line at test range 10.32°C to read the CWT at each flow — these three intersection points form Table 2 and are used in Cross Plot 2.



CROSS PLOT 1 [TEST 2 — POST FAN CHANGE] — CWT vs RANGE at Test WBT = 24.22°C · Test Range (red vertical line) = 10.32°C

TABLE 2 — CWT at Test Range = 10.32°C (Read from Cross Plot 1)

Flow Condition	Water Flow (m3/hr)	CWT at Test Range (°C)
90% Flow (Purple)	900	32.657
100% Flow (Green)	1000	33.513
110% Flow (Blue)	1100	34.332

STEP 3 — Calculation of Adjusted Water Flow (Q_{adj}) [TEST 2 — POST FAN CHANGE]

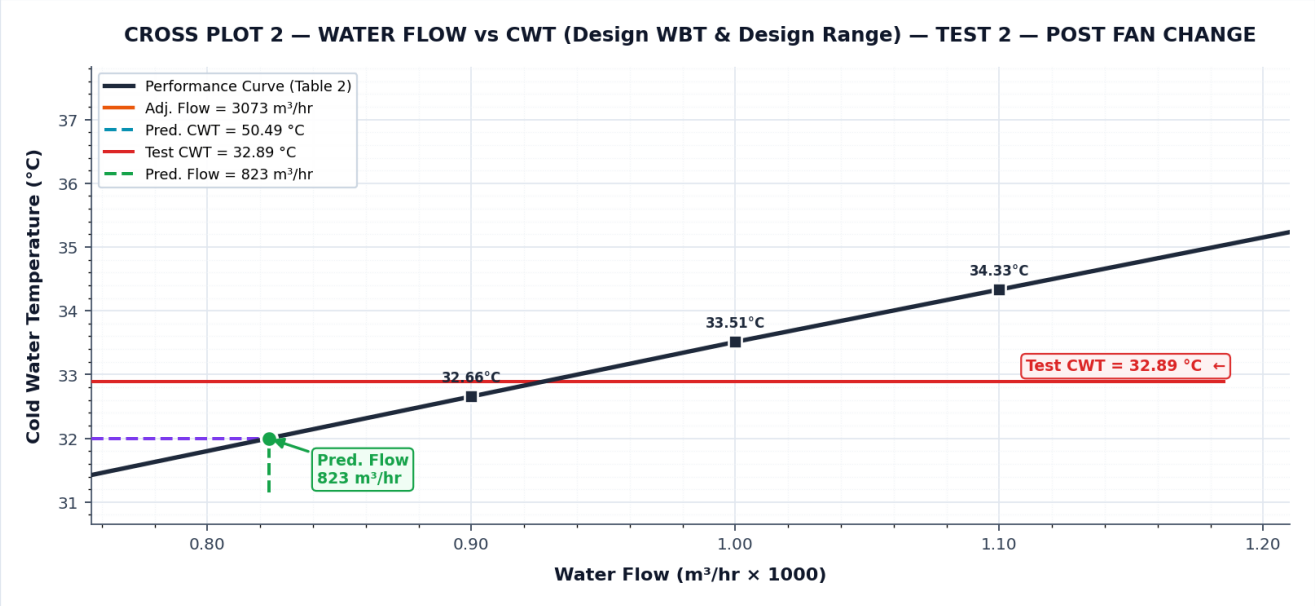
Correct the test water flow for fan power and air density differences between design and test conditions. The adjusted flow is then marked on Cross Plot 2.

Step 3 — Input Parameters & Adjusted Flow Calculation

Ref	Parameter	Value	Unit
a	**Fan Power Design (W_d)**	117	kW
b	**Fan Power Test (W_t)**	116.24	kW
c	**Hot Water Temperature (Test)**	43.21	Deg C
d	**Cold Water Temperature (Test)**	32.89	Deg C
e	**Range (Test)**	10.32	Deg K
f	**WBT (Test)**	24.22	Deg C
g	**Approach (Test)**	8.67	Deg C
h	**Test Water Flow (Q_{wt})**	3067	M3/hr
i	**Density Ratio (test / design)**	0.998737	—
$Q_{adj} = Q_{wt} \times (W_d/W_t)^{1/3} \times (\text{density ratio})^{1/3}$		**3072.59**	**m3/hr**

STEP 4 — Cross Plot 2 — Water Flow vs CWT [TEST 2 — POST FAN CHANGE]

Plot the three Table 2 points (Water Flow on X-axis, CWT on Y-axis). Mark Q_{adj} on the X-axis and project vertically to the curve to read the Predicted CWT. Project horizontally at Design CWT (32°C) to read the Predicted Water Flow.



CROSS PLOT 2 [TEST 2 — POST FAN CHANGE] — Water Flow vs CWT · Adjusted Flow = 3072.59 m3/hr · Predicted CWT = 50.49°C

FINAL ATC-105 RESULTS — TEST 2 — POST FAN CHANGE

STEP 5

Predicted CWT, Performance Shortfall and Capability — TEST 2 — POST FAN CHANGE

Mark the Adjusted Water Flow on the X-axis of Cross Plot 2 and project vertically to intersect the performance curve. Read the Predicted CWT horizontally from that point.

Ref	Item	Value	Unit
i	**Adjusted Water Flow (X-axis of Cross Plot 2)**	3072.59	m3/hr
ii	**Project vertical line → intersect curve → read CWT**	See Cross Plot 2 — orange vertical line at adjusted flow	
iii	**Predicted (Expected) Cold Water Temperature**	**50.49**	**Deg C**
iv	**Tolerance on CWT (Specification)**	0.00	Deg C
v	**Total Predicted CWT (incl. tolerance)**	**50.49**	**Deg C**
vi	**Test Cold Water Temperature (Recorded site data)**	32.89	Deg C
vii	**Predicted Water Flow (at Design CWT on Cross Plot 2)**	823.25	m3/hr

CONCLUSION — TEST 2 — POST FAN CHANGE

CWT Shortfall in Performance	Cooling Tower Capability
-17.60 °C	**373.2 %**
Test CWT (32.89°C) vs Predicted CWT (50.49°C)	= 3072.59 / 823.25 × 100

[TEST 2 — POST FAN CHANGE] — Shortfall = **-17.60 °C** · Capability = **373.2 %**

Test CWT is -17.60°C above the Predicted CWT. Capability = Adjusted Flow / Predicted Flow × 100 = 3072.59 / 823.25 × 100.

ATC-105 PERFORMANCE EVALUATION — TEST 3 — POST DISTRIBUTION CHANGE

Parameter	Unit	Design	Recorded Test
Water Flow Rate	M3/hr	1000	3680
Hot Water Temperature	Deg C	37	42.13
Cold Water Temperature	Deg C	32	32.4
Wet Bulb Temperature	Deg C	27	21.7
Range (HWT – CWT)	Deg C	5	9.73
Approach (CWT – WBT)	Deg C	5.0	10.7

STEP 1

Table 1 — CWT vs Range at Test WBT = 21.7°C · Design Range = 5°C

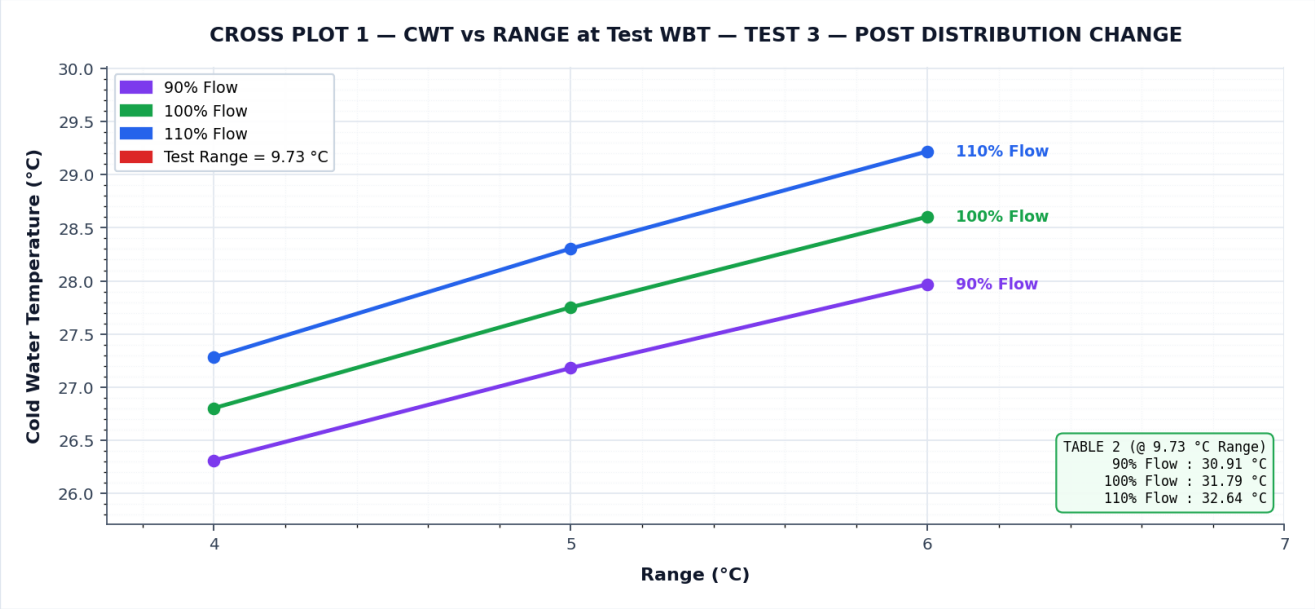
Performance curves at 90%, 100% and 110% water flow are evaluated at Test WBT = 21.7°C. CWT values at 80%, 100% and 120% of design range are computed for each flow condition.

Range %	Range °C	CWT 90% Flow (900)	CWT 100% Flow (1000)	CWT 110% Flow (1100)
80%	4 °C	26.31	26.80	27.28
100%	**5 °C**	**27.18**	**27.75**	**28.30**
120%	6 °C	27.97	28.61	29.22

STEP 2

Cross Plot 1 — CWT vs Range at Test WBT = 21.7°C [TEST 3 — POST DISTRIBUTION CHANGE]

CWT (Y-axis) vs Range (X-axis) at 90%, 100% and 110% water flow, evaluated at Test WBT = 21.7°C. Draw a vertical line at test range 9.73°C to read the CWT at each flow — these three intersection points form Table 2 and are used in Cross Plot 2.



CROSS PLOT 1 [TEST 3 — POST DISTRIBUTION CHANGE] — CWT vs RANGE at Test WBT = 21.7°C · Test Range (red vertical line) = 9.73°C

TABLE 2 — CWT at Test Range = 9.73°C (Read from Cross Plot 1)

Flow Condition	Water Flow (m3/hr)	CWT at Test Range (°C)
90% Flow (Purple)	900	30.907
100% Flow (Green)	1000	31.79
110% Flow (Blue)	1100	32.638

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STEP 3

Calculation of Adjusted Water Flow (Q_{adj}) [TEST 3 — POST DISTRIBUTION CHANGE]

Correct the test water flow for fan power and air density differences between design and test conditions. The adjusted flow is then marked on Cross Plot 2.

Step 3 — Input Parameters & Adjusted Flow Calculation

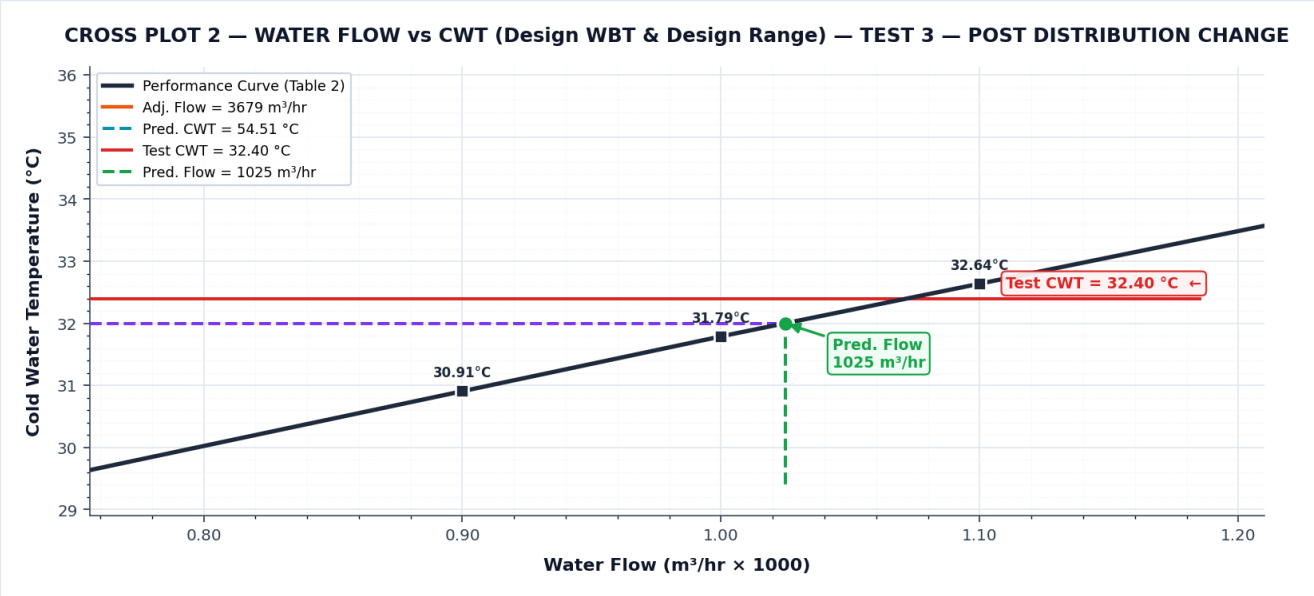
Ref	Parameter	Value	Unit
a	**Fan Power Design (W_d)**	117	kW
b	**Fan Power Test (W_t)**	117	kW
c	**Hot Water Temperature (Test)**	42.13	Deg C
d	**Cold Water Temperature (Test)**	32.4	Deg C
e	**Range (Test)**	9.73	Deg K
f	**WBT (Test)**	21.7	Deg C
g	**Approach (Test)**	10.7	Deg C

h	**Test Water Flow (Q_{wt})**	3680	M3/hr
i	**Density Ratio (test / design)**	0.999025	—
$Q_{adj} = Q_{wt} \times (W_d/W_t)^{1/3} \times (\text{density ratio})^{1/3}$		**3678.80**	**m3/hr**

STEP 4

Cross Plot 2 — Water Flow vs CWT [TEST 3 — POST DISTRIBUTION CHANGE]

Plot the three Table 2 points (Water Flow on X-axis, CWT on Y-axis). Mark Q_{adj} on the X-axis and project vertically to the curve to read the Predicted CWT. Project horizontally at Design CWT (32°C) to read the Predicted Water Flow.



CROSS PLOT 2 [TEST 3 — POST DISTRIBUTION CHANGE] — Water Flow vs CWT · Adjusted Flow = 3678.80 m3/hr · Predicted CWT = 54.51°C

FINAL ATC-105 RESULTS — TEST 3 — POST DISTRIBUTION CHANGE

STEP 5

Predicted CWT, Performance Shortfall and Capability — TEST 3 — POST DISTRIBUTION CHANGE

Mark the Adjusted Water Flow on the X-axis of Cross Plot 2 and project vertically to intersect the performance curve. Read the Predicted CWT horizontally from that point.

Ref	Item	Value	Unit
i	**Adjusted Water Flow (X-axis of Cross Plot 2)**	3678.80	m3/hr
ii	**Project vertical line → intersect curve → read CWT**	See Cross Plot 2 — orange vertical line at adjusted flow	
iii	**Predicted (Expected) Cold Water Temperature**	**54.51**	**Deg C**
iv	**Tolerance on CWT (Specification)**	0.00	Deg C

v	**Total Predicted CWT (incl. tolerance)**	**54.51**	**Deg C**
vi	**Test Cold Water Temperature (Recorded site data)**	32.4	Deg C
vii	**Predicted Water Flow (at Design CWT on Cross Plot 2)**	1024.76	m3/hr

CONCLUSION — TEST 3 — POST DISTRIBUTION CHANGE

CWT Shortfall in Performance	Cooling Tower Capability
-22.11 °C	**359 %**
Test CWT (32.4°C) vs Predicted CWT (54.51°C)	= 3678.80 / 1024.76 × 100

[TEST 3 — POST DISTRIBUTION CHANGE] — Shortfall = **-22.11 °C** · Capability = **359 %**

Test CWT is -22.11°C above the Predicted CWT. Capability = Adjusted Flow / Predicted Flow × 100 = 3678.80 / 1024.76 × 100.